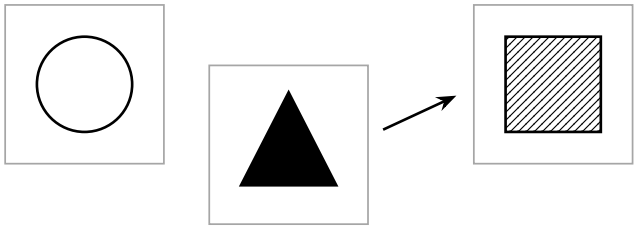
Use the nine tiles, one of each shape and fill. An allowed three uses three different tiles. Its shapes are all the same or all different. Its fills are also all the same or all different. Tile positions do not matter.



allowed three

shapes ○ △ □ all different

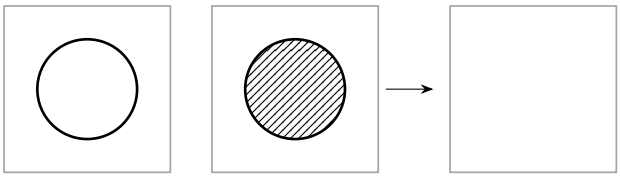
fills □ ■ ▨ all different

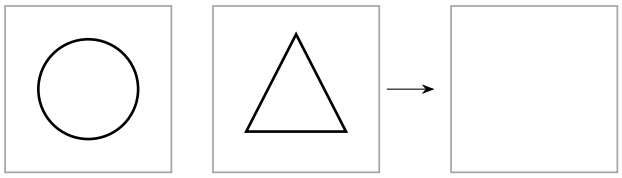
not allowed

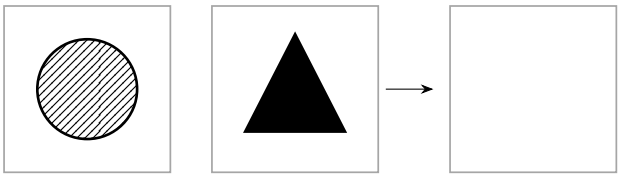
shapes ○ △ □ all different

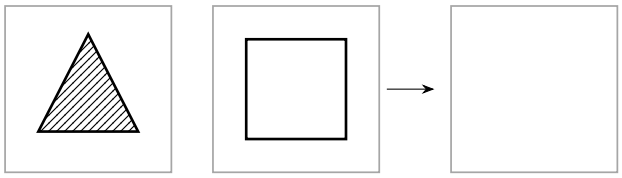
fills □ □ ■ two the same

Problem 1: Complete the pairs. Can two different tiles have two completing tiles, or none? Explain.

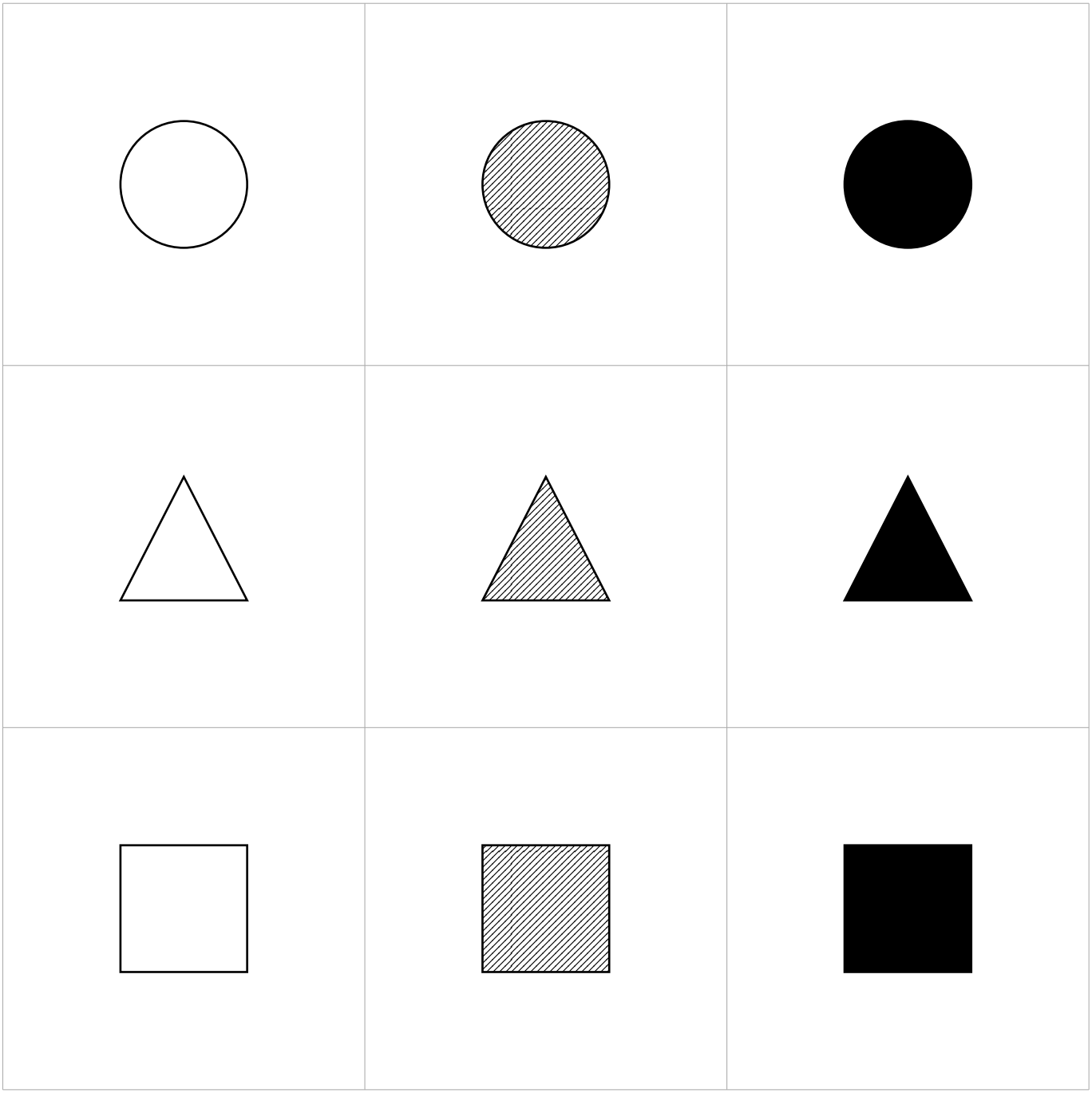




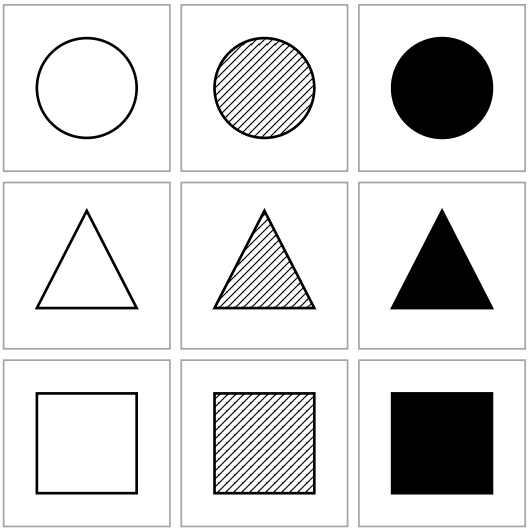


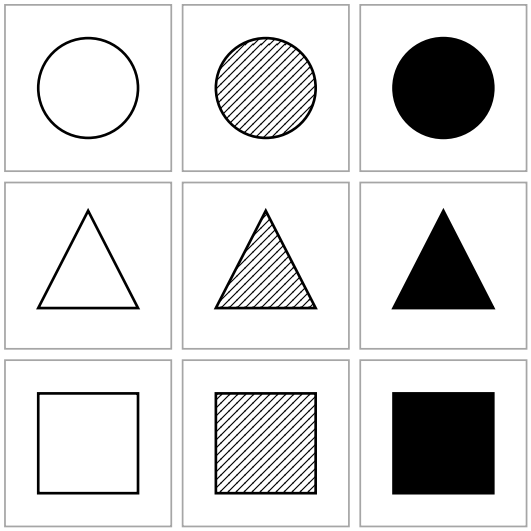


Problem 2: Choose as many tiles as possible without an allowed three among them. Explain why a larger collection is impossible.

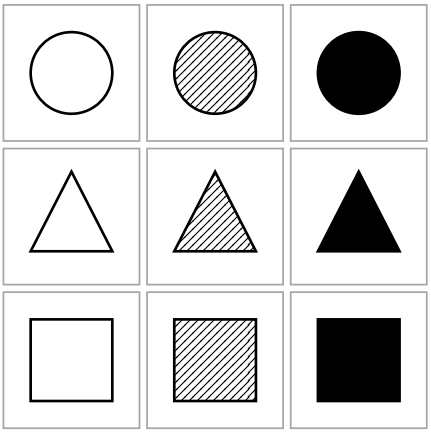


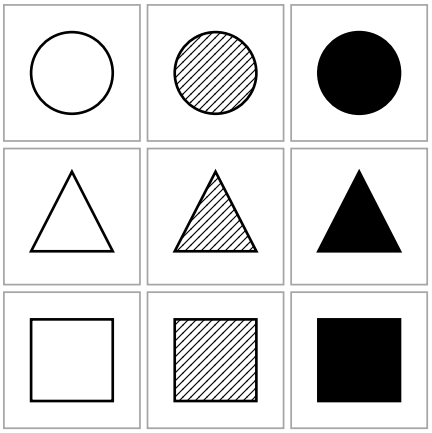
Problem 3: Two different allowed threes may share a tile. How many tiles can they share? Find as many allowed threes through one chosen tile as possible, and explain why you have them all.

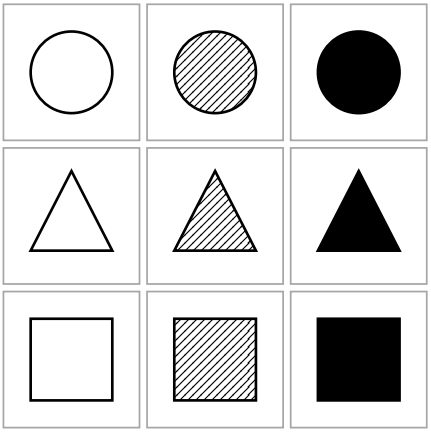


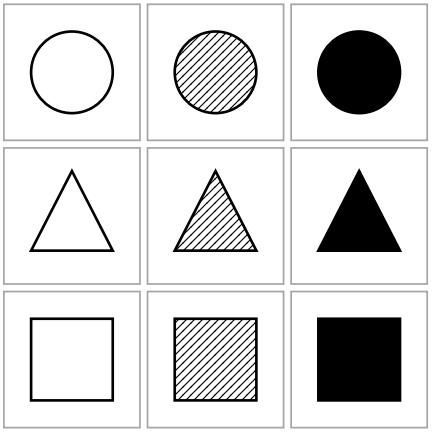


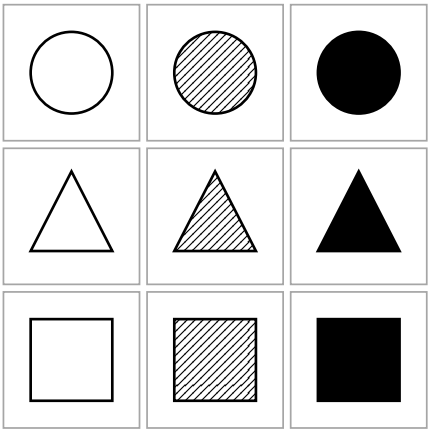
Problem 4: Split the nine tiles into three allowed threes. Find every split, and explain why your list is complete.

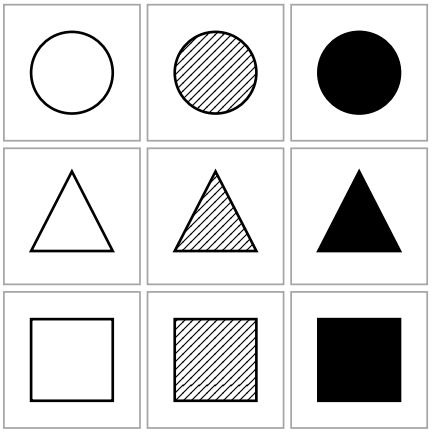












Problem 5: Start with no tiles chosen. Add one unused tile at a time without making an allowed three. Can different choices make you stop at different numbers of tiles? Explain why you can or cannot get stuck early.